Candidate Pseudolabel Learning: Enhancing VLMs with Unlabeled Data

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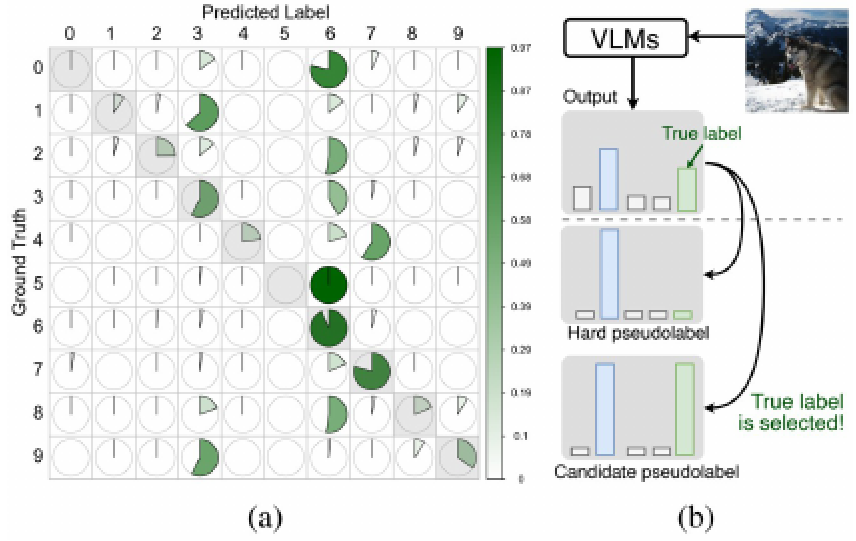
01 Zero-shot VLMs and the Labeling Bottleneck

- **Promise of zero-shot VLMs**, yet adaptation still needs labels
  - Labeled data is costly across diverse tasks and domains
- Unlabeled data is plentiful; pseudolabeling is attractive
  - Hard pseudolabels are often inaccurate and class-imbalanced
- Goal: adapt VLMs with minimal labels via **more reliable candidate pseudolabels**
  - Reduce reliance on high zero-shot accuracy and manual annotation

[Radford et al., 2021; Li et al., 2022; Yuan et al., 2021]
[Zhou et al., 2022a; Zhou et al., 2022b]

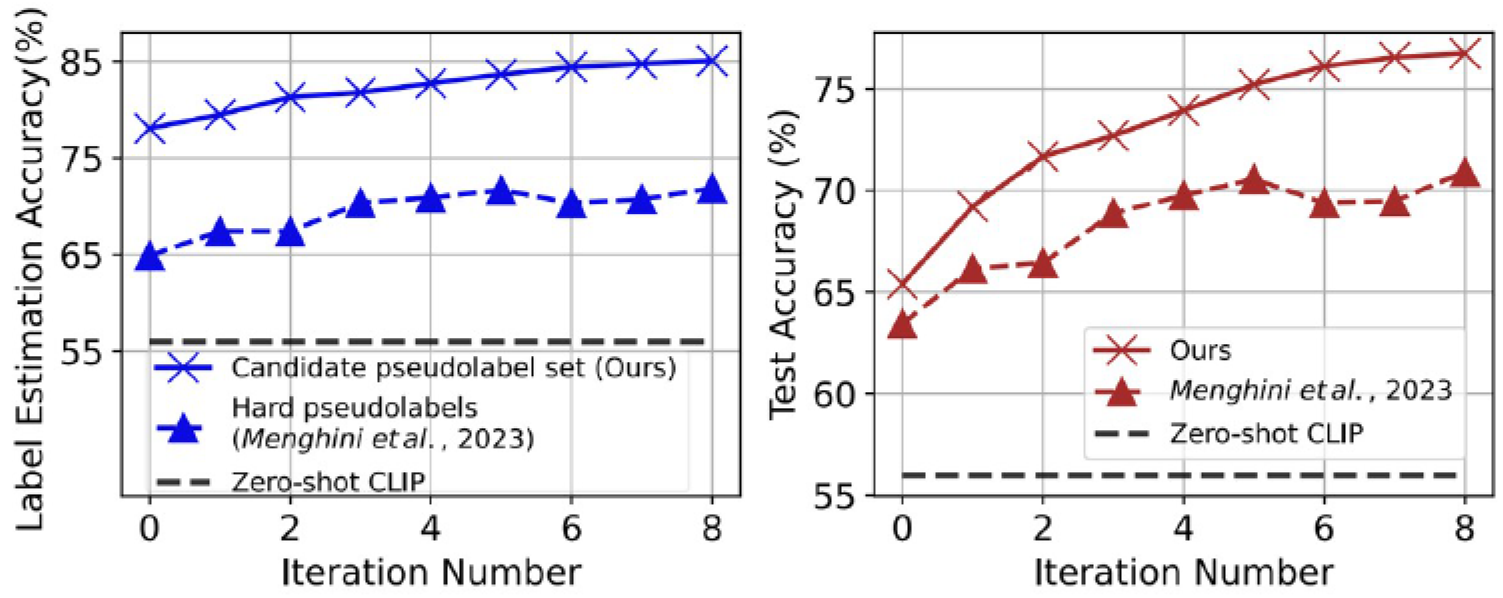
02 Hard Pseudolabel Pitfalls (EuroSAT)

- Hard pseudolabels skew toward dominant classes
  - EuroSAT shows many samples misassigned to a single class
- Fine-tuning on wrong and imbalanced labels degrades accuracy
  - Systematic errors propagate during training without correction
- Confusion analysis reveals **consistent bias across classes**
  - Motivates moving beyond top-one labels



[Helber et al., 2019; Radford et al., 2021; Menghini et al., 2023]

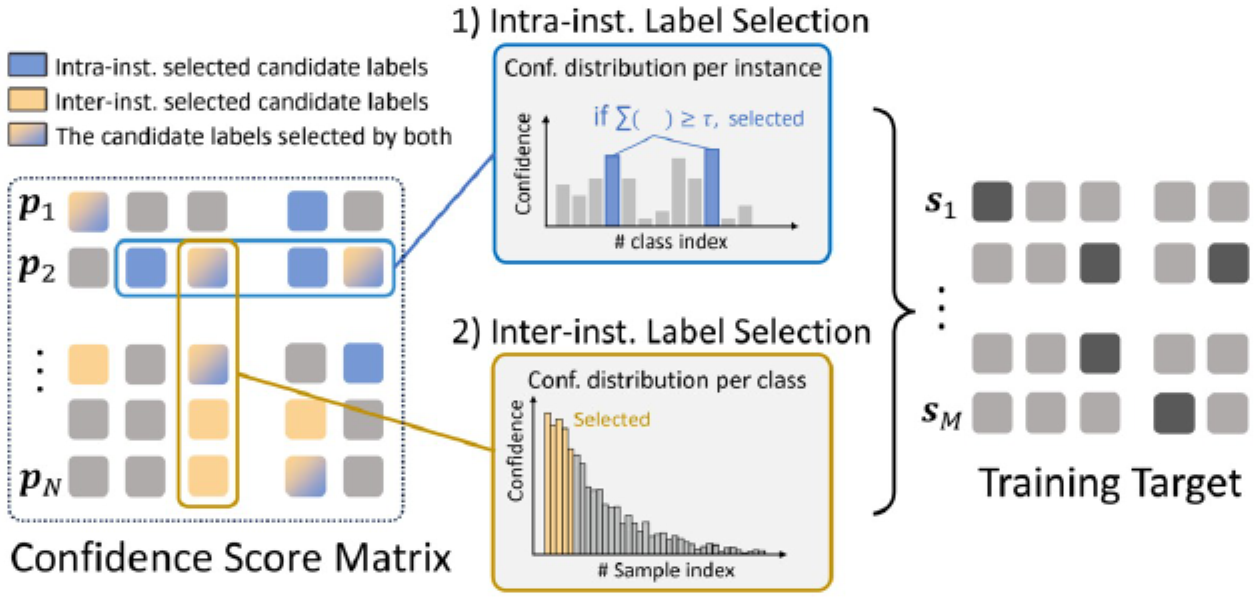
03 Candidate Sets: Intuition and Empirical Edge



- Use a set of top candidate labels per instance
  - Higher chance the true label is included than top-one
- **Empirically improves** label estimation and test accuracy
  - Performance rises across iterations on multiple datasets
- Treat all candidates equally to dilute class prior bias
  - Aligns with principles from partial-label learning

[Hossain and Kauranen, 2015; Li et al., 2023; Menghini et al., 2023]

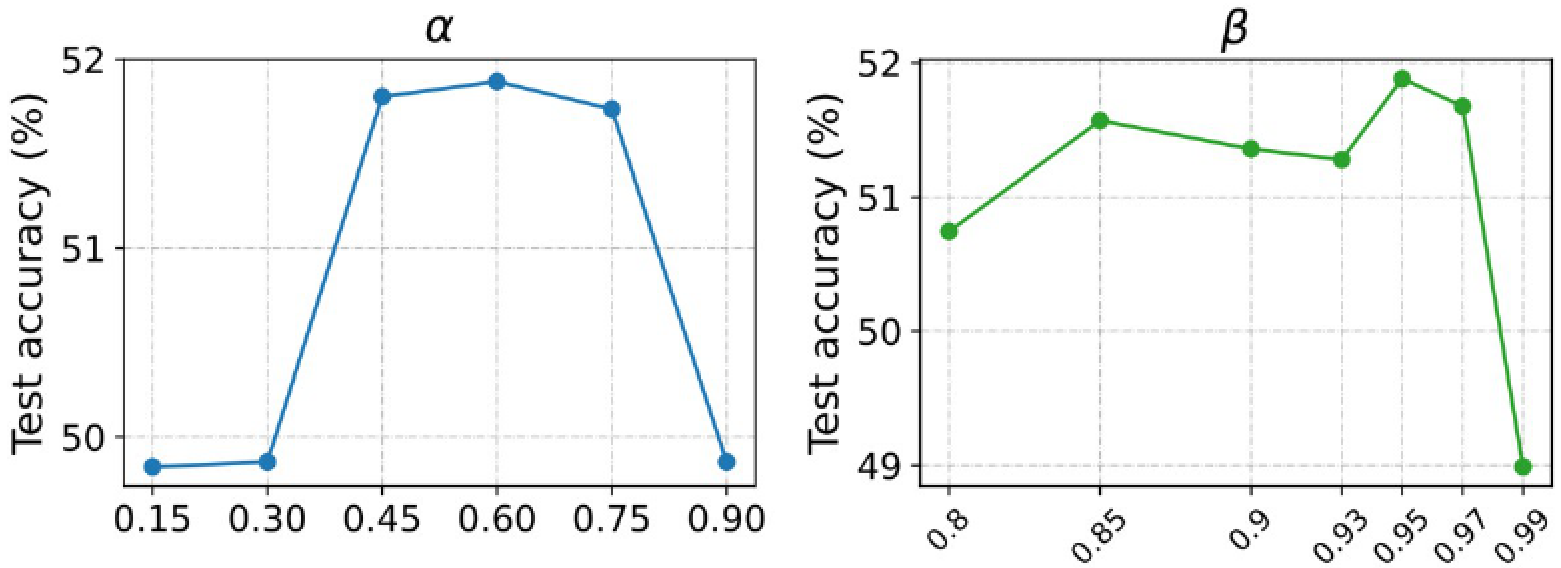
04 Workflow and Confidence Matrix



- Construct a confidence matrix over all unlabeled data
  - Each row stores per-class scores for one instance
- Generate candidate sets and train with partial-label losses
  - Iteratively refine candidates and the model parameters
- Balance classes via inter-instance selection and adapt the confidence threshold by a chosen quantile
  - Convert multiclass supervision into candidate-

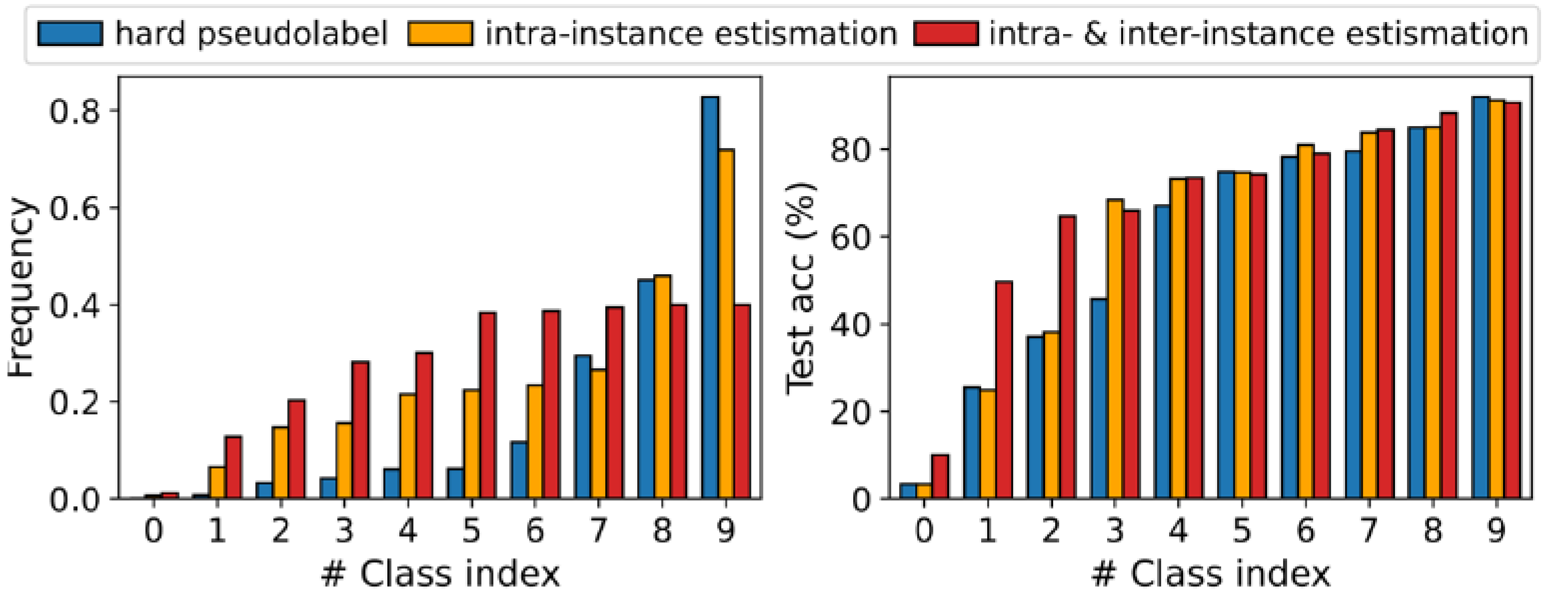
[Cour et al., 2011; Luo and Orabona, 2010]

05 Adaptive threshold via quantile of confidences



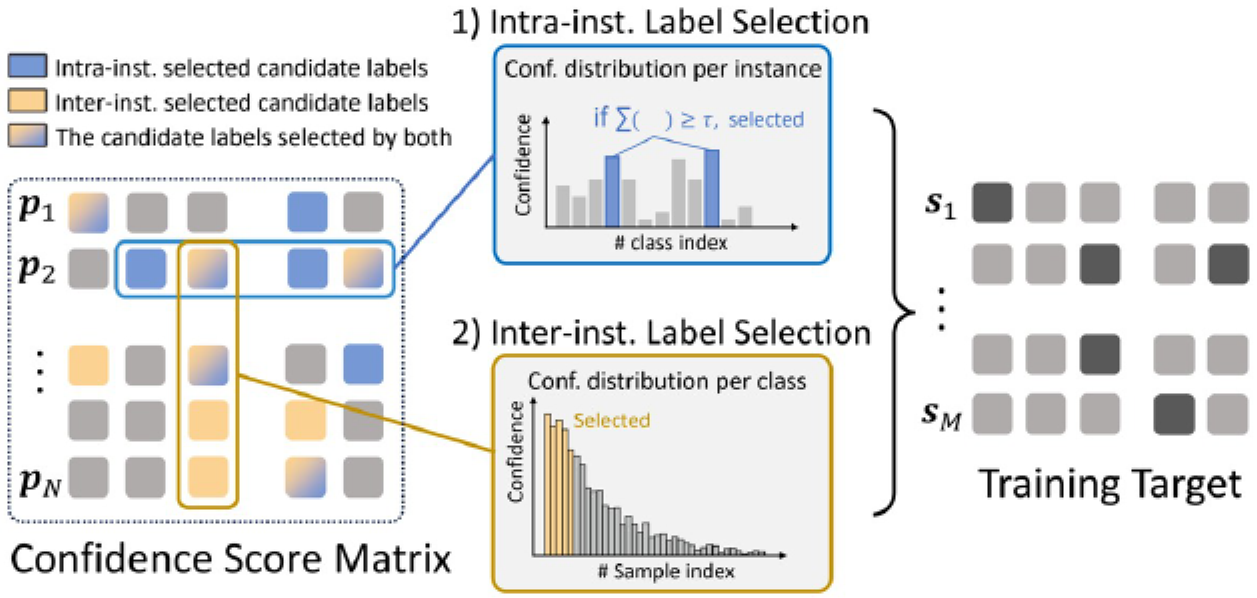
- Accumulate sorted class confidences until the sum reaches a threshold
  - Instance-specific candidate set size reflects difficulty
- The threshold adapts each iteration using a specified quantile of maximum confidences
  - Tracks improving calibration during training
- Limits ambiguity by enforcing comparable total confidence per instance
  - Reduces overly large candidate sets that hinder learning

06 Balancing with top-ratio and final intersection



- Per-class selection keeps top confident instances
  - Mitigates dominance of frequent or easy classes
- Final candidate set equals the intersection of intra- and inter-instance selections
  - Retains confident and balanced label options
- Tune the selection ratio to avoid empty sets and maintain coverage
  - **Stabilizes** iterative updates and class balance

07 Partial-label Losses and Iteration



- Adopt standard partial-label losses such as CC, RC, CAV, and LW
  - Flexible and effective across tasks
- Iterate: regenerate candidates and continue fine-tuning
  - Supports semi-supervised, unsupervised, and transductive zero-shot settings
- Construct binary target vectors from candidate sets and filter empty-candidate instances
  - Simple integration into existing training loops

[Feng et al., 2020; Wen et al., 2021; Zhang et al., 2021b]

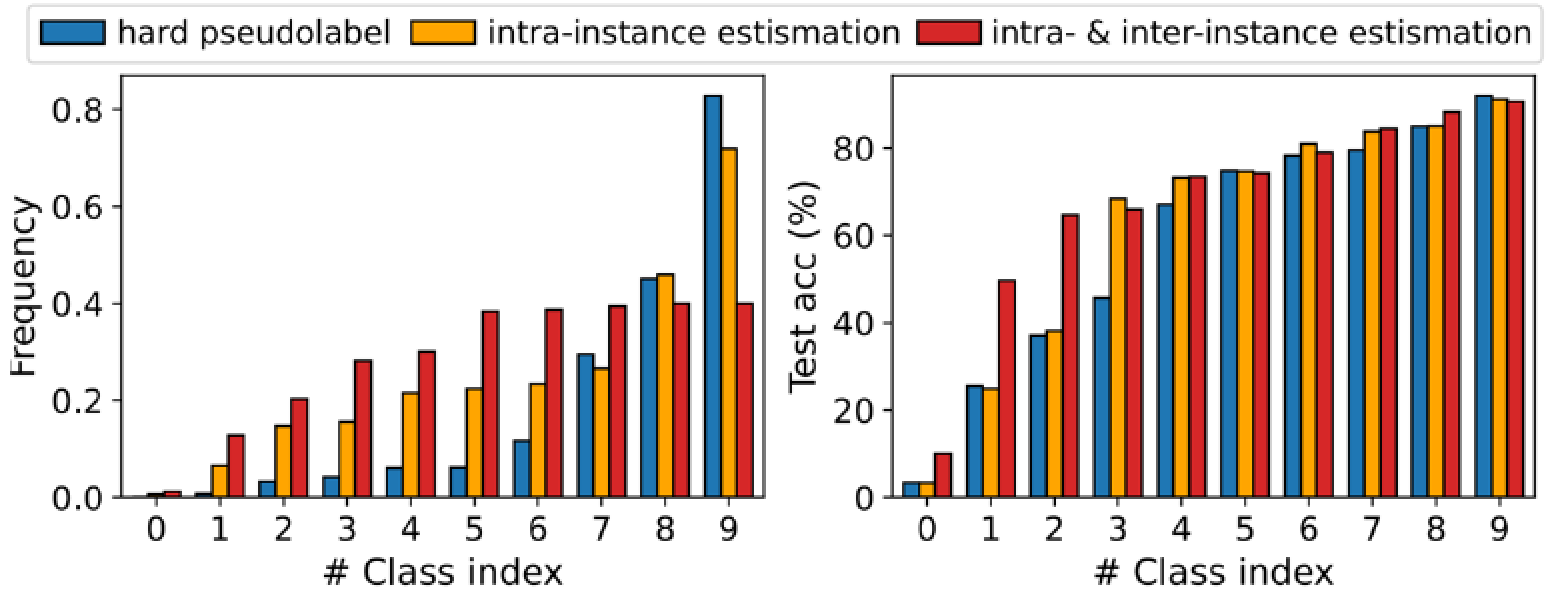
08 Main Results: Textual and Visual Prompts

Methods		Flowers102 SSL	Flowers102 UL	Flowers102 TRZSL	RESISC45 SSL	RESISC45 UL	RESISC45 TRZSL	DTD SSL	DTD UL	DTD TRZSL
Zero-shot CLIP		$63.67_{0.00}$		$63.40_{0.00}$	$54.48_{0.00}$		$54.46_{0.00}$	$43.24_{0.00}$		$43.45_{0.00}$
FPL	✗	$75.96_{0.74}$	$65.67_{0.23}$	$80.97_{0.00}$	$68.13_{0.55}$	$63.07_{0.38}$	$72.11_{0.00}$	$37.10_{5.45}$	$44.96_{0.55}$	$46.30_{0.03}$
CPL (Ours)	✗	$\mathbf{77.36}_{0.24}$	$\mathbf{70.01}_{0.21}$	$\mathbf{84.60}_{0.10}$	$\mathbf{71.73}_{0.57}$	$\mathbf{68.47}_{0.34}$	$\mathbf{72.16}_{0.26}$	$\mathbf{54.63}_{0.79}$	$\mathbf{48.92}_{0.17}$	$\mathbf{59.79}_{1.32}$
GRIP	✓	$83.60_{0.48}$	$69.84_{1.06}$	$86.26_{0.00}$	$74.11_{0.68}$	$70.55_{0.88}$	$81.07_{0.00}$	$56.07_{0.85}$	$46.09_{1.06}$	$65.30_{0.01}$
CPL (Ours)	✓	$\mathbf{89.66}_{0.36}$	$\mathbf{72.90}_{0.78}$	$\mathbf{87.35}_{0.76}$	$\mathbf{80.98}_{0.11}$	$\mathbf{77.39}_{0.44}$	$\mathbf{85.85}_{0.49}$	$\mathbf{61.21}_{0.56}$	$\mathbf{51.91}_{0.71}$	$\mathbf{68.00}_{0.34}$

Methods		CUB SSL	CUB UL	CUB TRZSL	EuroSAT SSL	EuroSAT UL	EuroSAT TRZSL	FGVCAircraft SSL	FGVCAircraft UL	FGVCAircraft TRZSL
Zero-shot CLIP		$51.82_{0.00}$		$51.57_{0.00}$	$32.88_{0.00}$		$30.54_{0.00}$	$17.58_{0.00}$		$17.86_{0.00}$
FPL	✗	$55.29_{0.59}$	$53.04_{0.53}$	$55.44_{0.20}$	$62.05_{1.64}$	$48.96_{1.49}$	$53.70_{26.87}$	$20.02_{0.77}$	$16.62_{0.67}$	$17.55_{0.37}$
CPL (Ours)	✗	$\mathbf{56.37}_{0.45}$	$\mathbf{54.18}_{0.05}$	$\mathbf{64.01}_{0.17}$	$\mathbf{64.84}_{2.15}$	$\mathbf{51.45}_{1.97}$	$\mathbf{54.03}_{2.27}$	$\mathbf{22.37}_{0.66}$	$\mathbf{18.90}_{0.20}$	$\mathbf{28.47}_{0.43}$
GRIP	✓	$56.65_{0.33}$	$51.42_{0.21}$	$59.48_{0.38}$	$58.66_{2.64}$	$57.21_{1.77}$	$92.33_{0.69}$	$16.98_{0.82}$	$15.22_{0.71}$	$26.08_{0.25}$
CPL (Ours)	✓	$\mathbf{58.53}_{0.24}$	$\mathbf{53.47}_{0.36}$	$\mathbf{66.20}_{0.50}$	$\mathbf{77.51}_{0.80}$	$\mathbf{67.26}_{0.47}$	$\mathbf{93.78}_{0.12}$	$\mathbf{22.48}_{0.63}$	$\mathbf{18.35}_{0.27}$	$\mathbf{30.86}_{0.70}$

- **CPL outperforms** hard pseudolabel baselines across datasets
  - Consistent gains with both textual and visual prompts
- Stronger improvements when zero-shot accuracy is weak
  - Notable on EuroSAT and DTD
- Works under semi-supervised, unsupervised, and transductive zero-shot paradigms
  - Scales with more unlabeled data

09 Robustness and Class Balance



- Less dependent on initial zero-shot accuracy
  - More resilient when initial predictions are weak
- Combined selections yield **more balanced training targets**
  - Reduces skew from dominant classes
- Equal-weight candidates avoid amplifying class priors
  - Iterative updates progressively denoise labels

[Huang et al., 2022; Menghini et al., 2023; Mirza et al., 2023]

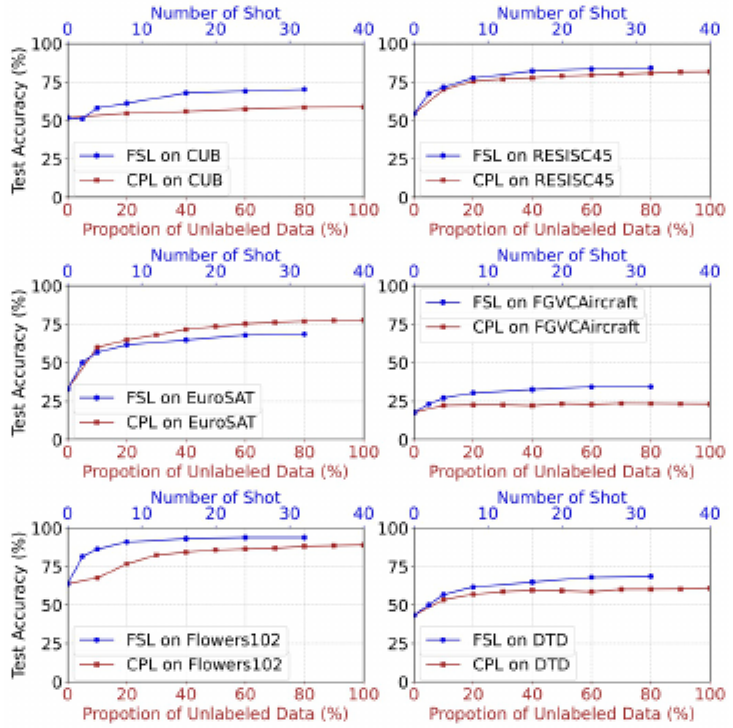
10 Loss Choices and Parameter-efficient Tuning

	Flowers-102	UCF-101	CIFAR-100	EuroSAT	DTD	CALTECH-101
CLIP	66.6	61.0	64.2	45.1	42.9	90.5
CLIP-PR	57.7	57.9	63.2	44.2	40.1	84.8
UPL	71.5	63.9	65.8	62.2	48.0	90.6
LaFTer	71.0	68.2	74.6	73.9	46.1	93.3
LaFTer + Ours	**76.7**	**71.0**	**77.3**	**82.2**	**56.3**	**93.4**

- CPL performs well with diverse partial-label losses
  - Competitive across CC, RC, CAV, LW
- Combining with LaFTer further boosts performance
  - Demonstrates plug-and-play versatility
- Soft cross-entropy is weaker than equal-weight candidate sets
  - Less effective at mitigating prior bias

[… al., 2020; Wen et al., 2021]
[Zhang et al., 2021b]

11 Unlabeled Proportion vs Few-shot Labeled



- Performance saturates as more unlabeled data is used
  - Diminishing returns beyond a point
- Small labeled sets can be more efficient on hard datasets
  - **CUB and FGVC-Aircraft** benefit from few-shot labels
- Trade off compute versus data quality when planning budgets
  - Hybrid strategies often work best

12 Imbalance and Encoder Scale

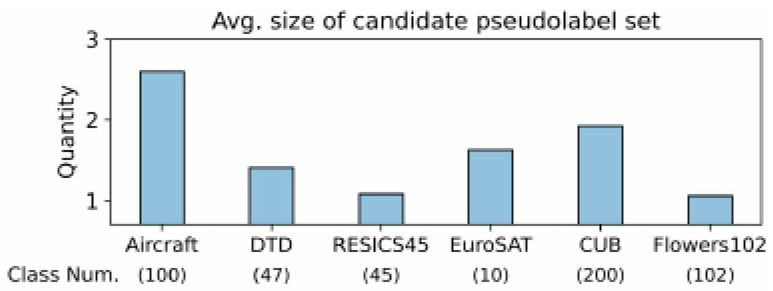
	Methods		SSL	UL	TRZSL
DTD	Zero-shot CLIP		$52.45_{0.00}$		$51.61_{0.00}$
	FPL	✗	$60.61_{1.56}$	$52.99_{0.43}$	$60.77_{0.54}$
	CPL	✗	$\mathbf{62.78}_{0.17}$	$\mathbf{57.23}_{0.19}$	$\mathbf{62.52}_{1.38}$
	GRIP	✓	$60.91_{0.00}$	$54.40_{0.00}$	$64.92_{0.00}$
	CPL	✓	$\mathbf{69.82}_{0.32}$	$\mathbf{57.20}_{0.45}$	$\mathbf{71.97}_{0.46}$
RESISC45	Zero-shot CLIP		$62.67_{0.00}$		$62.13_{0.00}$
	FPL	✗	$79.01_{0.55}$	$70.85_{0.66}$	$77.69_{0.83}$
	CPL	✗	$\mathbf{80.38}_{0.37}$	$\mathbf{76.01}_{0.19}$	$\mathbf{79.97}_{0.77}$
	GRIP	✓	$81.53_{0.00}$	$76.86_{0.00}$	$86.88_{0.00}$
	CPL	✓	$\mathbf{87.75}_{0.29}$	$\mathbf{80.88}_{0.86}$	$\mathbf{89.73}_{1.73}$
Flowers102	Zero-shot CLIP		$73.98_{0.00}$		$73.05_{0.00}$
	FPL	✗	$\mathbf{89.07}_{0.94}$	$77.81_{0.30}$	$91.84_{0.73}$
	CPL	✗	$88.37_{0.39}$	$\mathbf{82.98}_{0.14}$	$\mathbf{96.65}_{0.08}$
	GRIP	✓	$94.21_{0.00}$	$82.33_{0.00}$	$96.18_{0.00}$
	CPL	✓	$\mathbf{96.80}_{0.63}$	$\mathbf{83.94}_{0.69}$	$\mathbf{97.34}_{0.74}$

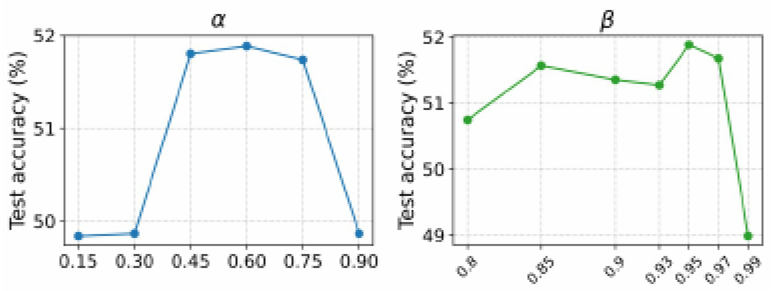
Methods	Balanced	Imbalanced δ=100	Imbalanced δ=50
LaFTer	74.64	65.63	66.59
CPL (w/o inter)	76.07	66.68	67.85
CPL	**77.32**	**67.70**	**69.65**

- CPL remains superior under class imbalance
  - Inter-instance selection further helps balance
- Gains persist with larger encoders
  - Scales to stronger backbones
- Stable improvements across learning paradigms with scale

[Zhou et al., 2020]

13 Hyperparameters and Set Size





- Two key knobs control candidate size and balance
  - Moderate quantile and high selection ratio work well in practice
- Average candidate set size is close to one
  - Keeps ambiguity low for most instances
- Avoid extreme selection ratios to prevent empty sets

14 Takeaways

- **Candidate pseudolabels** improve robustness and class balance
  - Stronger than hard pseudolabels across tasks
- Iterative refinement plus partial-label losses enables effective unlabeled adaptation
  - Simple to plug into existing prompt-tuning pipelines
- Versatile across prompts, learning paradigms, and encoders
  - Practical with sensible hyperparameter ranges

THANKS!